

**ADMINISTRATIVE INFORMATION****FMI No: 25398****FMI Title: VCT 06MW29.7 Patch**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems On-line Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

This FMI is **ONLY** applicable to **LightSpeed 7.x VCT** systems running **06MW29.7** software. **DO NOT** perform this FMI on any non-VCT systems running 06MW29.7 software.

Unit Tracked:**FMI Instructions Part No.: 5214715-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
On-Line Center
800-321-7937
Milwaukee, WI 53201, USA

GEMS - Europe

GEMS - E / ESC
BP34
F 78533 BUC CEDEX, FRANCE
Tel: (33 1) 39 20 00 07 or ESC
Fax: (33 1) 30 70 99 70

Asia Support Center
Phone: 81-426-56-0033
Fax: 81-426-48-2905

FMI No.: 25398

FMI Title: VCT 06MW29.7 Patch

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

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Revision History

Revision	Date	Reason for Change
1	3/20/2007	Initial release.
2	4/10/2007	Updated Appendix A to include the purpose for loading the software. Updated instructions to test the software in Appendix A.

Effectivity

This FMI affects all VCT systems with software revision 06MW29.7. The FMI should **NOT** be performed on any Discovery DVCT systems. The models affected in this FMI are:

Model#	Description
5124069	LIGHTSPEED VCT GANTRY
5124069-2	LIGHTSPEED VCT GANTRY
5124069-3	LIGHTSPEED VCT GANTRY
5124069-4	LIGHTSPEED VCT GANTRY
5124069-5	LIGHTSPEED VCT GANTRY
5124069-6	LIGHTSPEED VCT GANTRY
5126093	LIGHTSPEED VCT PRO32 / SELECT

NOTE: If your VCT system is running 06MW03.4 software revision, this FMI does not apply to you. **STOP** and close your dispatch as code **2 – FMI NOT REQUIRED**.

Purpose

The purpose of this FMI is to fix the following issues:

- **Scan Abort**

It was found on VCT systems with Halo detectors that scan aborts may occur during axial scans due to timing issues between the CANbus and the first view triggers.

- **FRDM Wizard**

It was found on VCT systems with Halo detectors that the FRDM Wizard may fail after changing the detector. The error message that is produced is similar to the following:

Host: wallvct Process: dastools.exe
File: spec_checker.c Line: 801
Internal software error.

- **Image Quality Issue**

It was found on VCT systems with Halo detectors that banding/image-to-image density change due to blocked reference channels & mis-application of correction data.

- **Dose**

It was discovered on VCT systems that any time an exam is paused, the following exam will not receive the proper mA feedback and report DLP will be off.

- **VCT Jalo Dose Negative**

It was found on VCT systems that when paused, the scanner can have an error in the timing sequence causing the firmware to report a negative dose.

Prerequisite FMIs

FMI Number	Title
FMI 25395	VCT GT Offset Issue

Related FMIs

None.

Obsoleted FMIs

None.

Furnished Materials

FMI 25398 kit p/n 5214715 contains:

Qty	Description	Part #
1	FMI Document	5214715-100
1	06MW29.7 Patch CD	5199298
1	Limited Access Patch CD	5192389
1	VCT Select GT Offset Issue Patch for 06MW29.7	5196620

Tool Required

None.

Personnel Requirements

1 FE for 3 hours labor and 1 hour of travel

Procedure

NOTE: It is recommended that you save systems state before beginning this procedure. At the end of this procedure, you will redo air cals and re-save system state.

Verify System Software Configuration

1. Open a Unix shell as ctuser.
2. Type: **swhwinfo -all <ENTER>**
3. The returning string should include the following software revision:

06MW29.7

NOTE: If this is not the software revision displayed, **STOP**. This FMI does not apply to your system.

Verify Prerequisite Patches Installed

1. As ctuser, type: **showprods | grep -i patch <ENTER>**
2. The output should contain:

Limited_Access_patch

Jalo_GT_5mm_Offset_Patch-1-1

If *Limited_Access_patch* is not installed, install CD P/N 5192389. See Appendix A for detailed installation instructions.

If *Jalo_GT_5mm_Offset_Patch-1-1* is not installed, install CD P/N 5196620. See Appendix B for detailed installation instructions.

NOTE: If FMI 25395 has not been completed previously and you install the *Jalo_GT_5mm_Offset_Patch-1-1* patch (P/N 5196620), close your dispatch for FMI 25395 – VCT GT Offset Issue as code 1. This software patch was also provided in that FMI.

If you need to load any of the patches, they should be loaded in the following order:

Order	Software Patch	CD Part #
1	Limited_Access_patch	5192389
2	Jalo_GT_5mm_Offset_Patch-1-1	5196620
3	06MW29.7_Patch	5199298

Load 06MW29.7 Software Patch

1. Shutdown all CT Applications.
2. Open a Unix shell.
3. Type: **su - <ENTER>**
4. Type password: **#bigguy <ENTER>**
5. Insert 06MW29.7 Patch CD, p/n 5199298 into the CT Host CDROM drive.
6. Type: **patch_install -c <ENTER>**

The output will be as follows:

These patches from cdrom will be installed:

06MW29.7_Patch

=====

I will install update 06MW29.7_Patch from CDROM, is this ok ? [y/n]

7. Type: **y <ENTER>**
8. Confirm that the patch has been successfully installed by typing:
 - a. Type: **showprods | grep -i patch <ENTER>**

The output will be as follows:

I 06MW29.7_Patch <date> Patch for 06MW29.7 ver 1 rel 1.1

9. Eject CD, type **eject <Enter>**
10. Type: **vrac_flash_update <ENTER>**

This tool will check the FLASH versions of the VRAC on all

IGs in the system. If any version is not up to date the

VRAC will be updated at this time. APPLICATIONS MUST NOT BE

RUNNING WHEN THIS STARTS. YOU MUST NOT ABORT THIS PROCEDURE

ONCE STARTED. Are you sure you want to continue?

enter 'y' to continue

11. Type: **y <ENTER>**
12. Type: **reboot <ENTER>**

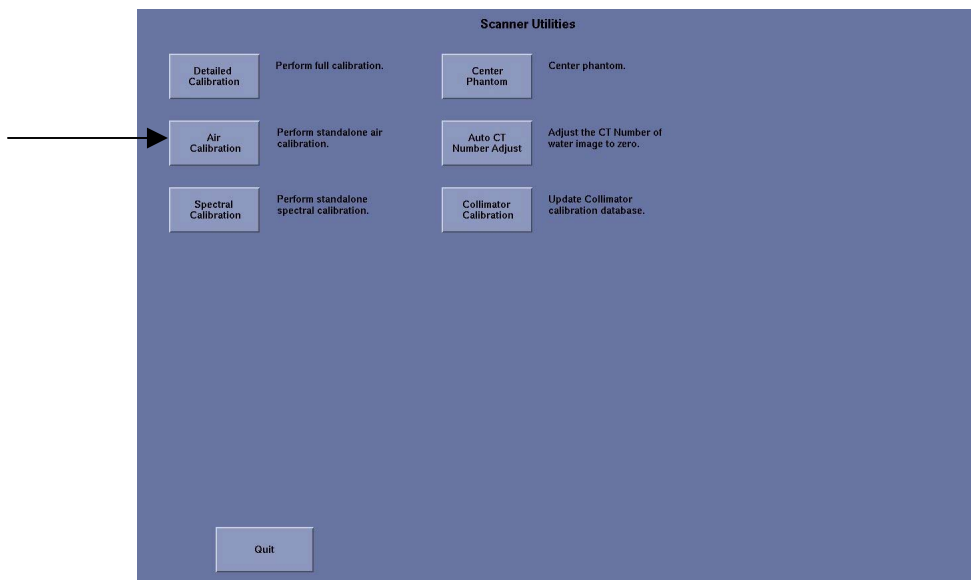
NOTE: If the following message box displays the prerequisite software was not installed prior to installation. Please review the ***Verify Prerequisite Patches Installed*** Section.



13. Wait for applications to start and then run **Flash Download Tool** located on the Common Service Desktop under Utilities.

Complete Calibration Update Process

1. Run Air Calibration in Scanner Utilities.



- a. Select Air Calibration.

- b. Standalone AIR Calibration screen appears. Select all techniques.

The image shows a screenshot of the 'Standalone AIR Calibration' screen. It features a grid of selection buttons for various parameters. On the right side of the screen, there is a text box stating: 'Air Calibration will be performed. Only Focal Spot is selectable.' At the bottom, there are 'Cancel' and 'Confirm' buttons.

Parameter	Options
KV	80KV, 100KV, 120KV, 140KV
SFOV	Small, Medium, Large
Focal Spot	Small, Large
Slice Collimation	2x0.625, 4x0.625, 8x0.625, 16x0.625, 32x0.625, 64x0.625

- c. Select Confirm.
2. Run FastCal to complete the calibration update process.

Finishing Up

1. Save system state to capture new Calibration Settings.
2. Perform a test scan to verify system functionality.
3. Remove FMI instructions from site.
4. Close your dispatch per normal regional procedure.

Appendix A – Limited Access Patch Instructions

1. Shutdown all CT Applications.
2. Open a Unix shell.
3. Type: **su - <ENTER>**
4. Type password: **#bigguy <ENTER>**
5. Insert Limited Access Patch CD, p/n 5192389 into the CT Host CDROM drive.
6. Type: **patch_install -c <ENTER>**

The output will be as follows:

These patches from cdrom will be installed:

Limited_Access_Patch

=====

I will install update Limited_Access_Patch from CDROM, is this ok ? [y/n]

7. Type: **y <ENTER>**
8. Confirm that the patch has been successfully installed by typing:
 - a. Type: **showprods | grep -i patch <ENTER>**

The output will be as follows:

Limited_Access_Patch

9. Eject CD, type **eject <Enter>**

Appendix B – VCT GT Offset Correction Software

It was found that a potential inaccuracy can occur in the positional display when the manual cradle release on VT 1700 and VT 2000 patient tables is used inappropriately during Radiation Therapy simulation procedures.

The manual cradle release feature is intended for emergency egress and small movements during CT fluoroscopy procedures. In rare instances, if the cradle release is used to move a patient out of the gantry, instead of using the table control buttons as specified in the operator manual, a discrepancy of up to 6mm between the numerical display on the gantry and the actual position of the table can be introduced.

1. Shutdown Applications.
2. Open a Unix shell as ctuser.
3. Type: **touch /usr/g/fw/tgpFtp /usr/g/fw/orpFtp <ENTER>**
4. Type: **cd /usr/g/fw <ENTER>**
5. Type: **ls *Ftp <ENTER>**

Verify that **tgpFtp** and **orpFtp** exist. (If those two files are not found or misspelled, repeat the step 3.)

6. Install patch CD p/n 5196620 in the Host computer.
 - a. Type: **su - <ENTER>**
 - b. Type: **patch_install -c <ENTER>**
 - c. Type: **y <ENTER>**

The load will take about 30 seconds and the following output will be displayed:

JALO_GT_5mm_Offset_Patch has been successfully installed.

7. Verify software loaded correctly.
 - a. Type: **showprods | grep -i patch <ENTER>**

The output should be:

Limited_Access_patch

JALO_GT_5mm_Offset_Patch-1-1

8. Eject CD, type **eject <ENTER>**
9. Type: **reboot <ENTER>**
10. Run Flash download tool (Query and Update) located on the Common Service Desktop under Utilities.
11. Perform **Verify Patch Functionality**.
 - a. Move the table to home position.

- i. Lift the table to the highest position and press “Cradle Out” to move the cradle away from the gantry to its electrical limitation.

- b. Unlatch the cradle and pull the cradle out away from the gantry to its mechanical limitation until the cradle can't be pulled anymore.
- c. Press the "Landmark" button, since the downloaded firmware is disables the setting of the landmark is this special condition, you may observe that the landmark can not be set then.

The error log should be the following; the "Cradle position" and "IMS position" should refer to

```
Exception Class : Secondary   Severity : Pri/Soft
File : uif_table_tilt_mgr_gt.c   Line# : 6057
Function : No System Function Reported
Scan Type : None/Unknown Scan: 0/0/0
Landmark button is ignored because it's pressed in invalid location.
Press <Cradle-IN> and press Landmark again.
Cradle position -4590.
IMS position 0.
```

actual values and may not be the same as follows.

12. Perform a test scan to verify system functionality.

Appendix C – Software Uninstall Instructions

1. Shutdown all CT Applications.
2. Open a Unix shell.
3. Type: **su - <ENTER>**
4. Type password: **#bigguy <ENTER>**
5. Insert 06MW29.7 Patch CD, p/n 5199298 into the CT Host CDROM drive.

NOTE: Be careful to only remove the software intended as others may will be loaded on this system.

Other patches may be loaded other than those listed in the following steps.

6. Type: **patch_install -u <ENTER>**

There are following pathes installed:

Limited_Access_patch

JALO_GT_5mm_Offset_Patch

06MW29.7_Patch

=====

Uninstall update Limited_Access_patch, is this ok? [y/n]

7. Type: **n <ENTER>**

Exiting on user command

Uninstall update JALO_GT_5mm_Offset_Patch, is this ok? [y/n]

8. Type: **n <ENTER>**

Exiting on user command

Uninstall update 06MW29.7_Patch, is this ok? [y/n]

9. Type: **y <ENTER>**

Erasing: 06MW29.7_Patch 1/1

/bin/mv: cannot stat `/usr/local/gemsvrac/vrac2_flash/pbc1113-bpp1015.elf.pre_06MW29.7_Patch-1-1': No such file or directory

/bin/mv: cannot stat `/usr/local/gemsvrac/vrac2_flash/pbc115-bpp32.elf.pre_06MW29.7_Patch-1-1': No such file or directory

Erased: 06MW29.7_Patch 1-1.i386

06MW29.7_Patch has been uninstalled successfully.

=====

=

10. Confirm that the patch has been successfully installed by typing:
 - a. Type: **showprods | grep -i patch <ENTER>**
The output will be similar to the following:
Limited_Access_patch
Jalo_GT_5mm_Offset_Patch-1-1
11. Eject CD, type **eject <Enter>**
12. Type: **vrac_flash_update <ENTER>**
*This tool will check the FLASH versions of the VRAC on all
IGs in the system. If any version is not up to date the
VRAC will be updated at this time. APPLICATIONS MUST NOT BE
RUNNING WHEN THIS STARTS. YOU MUST NOT ABORT THIS PROCEDURE
ONCE STARTED. Are you sure you want to continue?
enter 'y' to continue*
13. Type: **y <ENTER>**
14. Type: **reboot <ENTER>**
15. Wait for applications to start and then run **Flash Download Tool** located on the Common Service Desktop under Utilities.
16. Run Air Calibration in Scanner Utilities.
 - a. Select Air Calibration.
 - b. Standalone AIR Calibration screen appears. Select all techniques.
 - c. Select Confirm.
17. Run FastCal to complete the calibration update process.
18. Save system state to capture new Calibration Settings.